



Developmental Barriers and Transition Pathways for Teachers' Digital–Intellectual Thinking: A Cognitive Ecology Perspective



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Abstract: Educational digital transformation has been elevated to a national strategic priority, with the development of teachers' digital–intellectual thinking emerging as a critical determinant of transformation effectiveness. In practice, however, teachers commonly encounter a dilemma characterized by “learning without application” and “superficial application” when engaging with intelligent technologies. Although teachers perceive themselves as competent in evaluating online information, their actual performance frequently falls short of effective implementation. Furthermore, while self-efficacy in information literacy enhances teaching engagement, the process of technology integration is accompanied by complex emotional experiences. Existing research predominantly examines singular dimensions such as teachers' information literacy or technology acceptance, thereby failing to uncover the systemic mechanisms underlying retardant effects. By introducing cognitive ecology theory and employing literature analysis, questionnaire surveys, and in-depth interviews, the retardant effects impeding the development of teachers' digital–intellectual thinking were systematically diagnosed. The findings indicate that these retardant effects manifest primarily in four forms: cognitive inertia fixation, data–meaning construction fracture, human–machine collaborative perception misalignment, and professional identity narrative disruption. These manifestations are rooted in a structural imbalance among individual cognitive schemas, technological environment provision, and community-of-practice ecologies. Accordingly, an ecological transition pathway was designed, encompassing “cognitive restructuring—environmental enabling—ecological synergy.” This study deepens the understanding of the developmental patterns of teachers' digital–intellectual thinking, extends the applicability of cognitive ecology theory within the field of educational digitalization, and provides actionable diagnostic tools and practical implementation strategies for teacher training and the construction of school-based digital–intellectual ecologies.

Keywords: Cognitive ecology theory; Teachers' digital–intellectual thinking; Retardant effects; Transition pathways; Educational digitalization

1. Introduction

The underlying logic of the educational system is being reshaped by educational digital transformation (Li & Xue, 2025; Wang et al., 2025). In 2022, the teacher digital literacy industry standard was promulgated by the Ministry of Education, which delineated a standard framework for teachers' digital literacy in the new era across five dimensions: digital awareness, digital technology knowledge and skills, digital application, digital social responsibility, and professional development. For the first time, the 20th National Congress of the Communist Party of China proposed the initiative to “advance educational digitalization and build a learning society and a learning powerhouse for lifelong learning for all”, thereby endowing educational digitalization with a renewed mission in the construction of a modern socialist country. Against this backdrop, the development of teachers' digital–intellectual thinking has emerged as a critical determinant of transformation effectiveness.

In practice, however, teachers commonly encounter a dilemma characterized by “learning without application” and “superficial application” when engaging with intelligent technologies (Omar et al., 2025). Although teachers

perceive themselves as competent in evaluating online information, their actual performance evaluations have revealed no significant improvement, with a notable discrepancy between perceived competence and actual performance (Pavlounis et al., 2025). Teachers' digital literacy exhibits a structural feature of "stronger digital awareness and digital ethics, weaker digital knowledge and digital competence"; significant differences in digital literacy levels are observed across teaching seniority and school segments; and the development of dimensions such as digital awareness, digital technology knowledge and competence, digital application, and professional development remains unbalanced (Hamurcu, 2026; Liu et al., 2025). This retardant phenomenon in the development of digital–intellectual thinking reflects a more profound issue: the renewal of teachers' cognitive systems is not a mere process of skill acquisition, but is deeply embedded in the complex interplay among individual cognitive habits, technological environment provision, and organizational cultural ecology (Tam, 2025; Tsagari & Armostis, 2025). It has been noted that educational digitalization imposes unprecedented pressure and challenges on teachers' professional growth; in the process of higher education digital transformation, university teachers, constrained by instrumental rationality dependence, insufficient digital literacy, and identity crises, have been trapped in a situation of "forced passive transformation" (Garrison & Oddone, 2026). However, existing research predominantly examines singular dimensions such as teachers' information literacy or technology acceptance, thereby failing to uncover the systemic mechanisms underlying retardant effects, and remains deficient in an integrative theoretical framework to guide the design of pathways for breaking through these predicaments (Chen et al., 2025a).

Cognitive ecology theory emphasizes that cognitive activities occur as a dynamic coupling of agents, environments, and technologies, thereby providing a fitting theoretical lens for understanding the dilemmas in the development of teachers' digital–intellectual thinking. The formation mechanisms of technological cognition in educational contexts have been examined from the perspective of cognitive ecology by certain scholars; learning pathways for teachers' deep learning have also been explored from the standpoint of ecosystem theory; and the enhancement of teachers' digital literacy has been scrutinized from an educational ecology perspective, with the argument that certain educational ecological principles ought to be followed (Çelik Çoban & Büyükdere Atadag, 2026). In view of this, cognitive ecology theory was introduced into research on teacher professional development. Based on a systematic diagnosis of the manifestations and roots of retardant effects in the development of teachers' digital–intellectual thinking, an ecological transition pathway of "cognitive restructuring—environmental enabling—ecological synergy" was designed for exploration—wherein cognitive restructuring is oriented toward the renewal of teachers' data literacy and attitudes toward intelligent technologies (Athanasopoulos et al., 2026); environmental enabling emphasizes the shift of intelligent platforms from functional aggregation to cognition-adaptive design (Sari & El Islami, 2025); and ecological synergy points to the systematic adjustment of teaching research institutions and evaluation mechanisms (Chen et al., 2025b; Clarke et al., 2025; Xiao et al., 2026)—so as to provide theoretical references and practical implications for teachers to break through cognitive barriers and achieve professional growth in the digital intelligence era.

2. Literature Review and Theoretical Framework

2.1 Connotations and Research Progress of Teachers' Digital–Intellectual Thinking

Digital–intellectual thinking is conceptualized as an advanced cognitive capacity possessed by teachers within digital and intelligent educational contexts, through which data literacy, artificial intelligence literacy, and critical thinking are integrated for the understanding, reasoning, and decision-making of educational and instructional problems, with an emphasis on cognitive restructuring in human–machine collaboration. The concept of "technology-enhanced cognition" has been proposed in related studies, suggesting that intelligent technologies are capable of extending and amplifying teachers' cognitive functions. With respect to teachers' information literacy, it has been argued that teachers must first acquire the necessary media literacy skills and pedagogical knowledge before effective integration into teaching can be achieved. Empirical studies, however, have revealed that although teachers report increased confidence following training participation, performance assessments indicate that skills have not improved correspondingly, with a significant discrepancy observed between perceived competence and demonstrated ability. The evaluation index system for information literacy of master of education students constructed by Chen et al. (2025b) similarly indicated that their information awareness and ethical levels were relatively strong, whereas information knowledge and skills, as well as information behavior and innovation, remained comparatively weak. Xie et al. (2026) identified information literacy self-efficacy as the primary factor promoting engagement in information-based teaching, with emotions serving as parallel mediators in this process. The comparative study conducted by Aillerie et al. (2025) in France and Peru revealed the significant impact of teacher training and educational policies on the implementation of media and information literacy. Further research found that urban–rural disparities, gender differences, and experience variations played notable roles in teachers' academic information interactions.

In summary, the majority of existing studies remain confined to the descriptive level of technological functions,

and lack a systematic explanation for the dynamic generation and developmental stagnation mechanisms of digital–intellectual thinking. How to accurately assess and effectively enhance teachers’ digital literacy and competencies continues to attract widespread attention, and a more integrative theoretical perspective is urgently needed.

2.2 Retardant Effects in Teacher Professional Development

Existing studies on retardant effects have predominantly been approached from single dimensions. At the individual level, cognitive inertia, technology anxiety, and low self-efficacy have been identified as major internal factors; unprecedented pressure and challenges have been imposed on teachers by educational digitalization, and the subjectivity of university teachers has been obscured under the combined effects of technological acceleration, professional inertia, and institutional discipline (Aillerie et al., 2025). At the organizational level, administrative mandate-driven implementation, the disconnect between training and practice, and lagging evaluation mechanisms have been repeatedly cited; significant differences in digital literacy have been observed among teachers with varying years of service, professional titles, and subject backgrounds, and the “three-not” dilemma (not being able, not being willing, and not being daring) persists. At the institutional level, the homogenization of policy supply and uneven resource allocation have exacerbated regional disparities, and the enhancement of digital literacy among rural and township teachers has been confronted with practical predicaments such as cultural erosion and supply-demand obstruction.

Although such studies have revealed multiple manifestations of retardant effects, these manifestations have rarely been examined within a unified framework that investigates their interactive relationships. While certain scholars have explored pathways for teachers’ deep learning or constructed three-dimensional analytical models based on ecosystem theory (Xiao et al., 2026), the question of “why significant differentiation in teacher development occurs under identical policy and technological conditions” has not been adequately addressed, and an ecologically holistic solution remains lacking.

2.3 Cognitive Ecology Theory

Cognitive ecology theory originates from ecological psychology and distributed cognition research, whereby cognition is understood as an emergent phenomenon within an ecosystem constituted by cognitive agents, artifacts, social interactions, and cultural practices, with an emphasis on the dynamic matching of “affordance” and “effectiveness”. In educational research, this theory has been employed to examine the formation mechanisms of artificial intelligence cognitive dependency and the intrinsic mechanisms of digital empowerment (Lin, 2026). Its introduction into research on teachers’ digital–intellectual thinking entails that technological platforms, teaching research institutions, peer networks, and school cultures are to be regarded as constituent elements of the cognitive ecology, and that retardant effects and transitions are to be examined from the three-dimensional interactive perspective of “individual–technology–community.”

On this basis, a three-dimensional analytical framework of “cognitive agent–technological environment–community of practice” is constructed. The cognitive agent dimension focuses on teachers’ knowledge structures, cognitive schemas, and professional agency, self-efficacy and emotions jointly influence engagement in information-based teaching. The community of practice dimension includes teaching research collaboration, inter-school networks, and institutional incentives—as demonstrated by Aillerie et al. (2025), training and policies exert significant influences on the implementation of media and information literacy.

The dynamic logic underlying the three-dimensional framework lies in the following: whether technological “affordances” can be perceived and utilized by teachers depends on whether existing schemas of the cognitive agent are compatible with them—sociodemographic variables have been found to significantly influence the level of information literacy enhancement. The community of practice, in turn, plays a moderating role between the two: strongly collaborative communities mediate the perception and meaning-making of technological affordances, whereas weakly collaborative or homogenized communities reinforce cognitive inertia, resulting in technological affordances that are “available but not adopted.” Retardant effects are essentially the self-locking of the system when “mismatches” occur across the three dimensions. Within this framework, the cognitive agent dimension focuses on teachers’ knowledge structures, cognitive schemas, and professional agency; the technological environment dimension encompasses the architecture of intelligent teaching platforms, data feedback mechanisms, and adaptive push design; and the community of practice dimension includes teaching research collaboration, inter-school networks, and institutional incentives. The three dimensions interact with one another in pairwise fashion, constituting a dynamically evolving cognitive ecosystem, in which mismatch in any single dimension may result in holistic developmental obstruction. This framework provides a theoretical anchor for subsequent empirical analysis and pathway design.

3. Research Methods

3.1 Research Design

A sequential explanatory mixed-methods design was adopted for this study, which was implemented in three phases. In the first phase, a cross-sectional questionnaire survey was administered, aimed at diagnosing, on a relatively large scale, the current status of retardant effects and their influencing factors in the development of teachers' digital–intellectual thinking; the identification checklist for retardant phenomena across the dimensional components is presented in Table 1. In the second phase, a longitudinal multi-case qualitative study was conducted, through which the generative mechanisms of retardant effects and the micro-processes of ecological transition were revealed via continuous tracking of typical teachers and their respective schools. In the third phase, a quasi-experimental action research was carried out, in which the transition pathway of “cognitive restructuring—environmental enabling—ecological synergy” was implemented in selected schools to verify its effectiveness. Cognitive ecology theory was applied throughout the entire process as a meta-theory for data analysis and framework construction.

Table 1. Identification checklist for retardant phenomena across dimensional components

Retardant Phenomenon	Behavioral Signals	Typical Utterances
Learning without application	Sharp decline in account activity	“I learned it, but I don’t know how to use it.”
Wait-and-see attitude	Attention paid to others’ practices while remaining inactive	“I’ll wait until it matures.”
Identity narrowing	Treatment as someone else’s responsibility	“I’m a subject X teacher; I can’t handle technology.”
Superficial application	Use of only the simplest functions	“Isn’t PowerPoint sufficient?”
Low self-efficacy	Abandonment after one or two attempts	“No improvement in student performance was seen.”

The cognitive agent dimension corresponded to the four scales of cognitive inertia, meaning-making, collaborative perception, and identity narrative in the questionnaire, as well as interview narratives, through which knowledge structures, schema updating, and professional agency were measured. The technological environment dimension relied on platform logs and classroom observations, in which the degree of technological affordance matching was captured through indicators such as the frequency and types of function calls, operation depth, modes of system recommendation adoption, and degree of algorithmic understanding. The community of practice dimension was examined through interviews, teaching research records, and institutional documents, in which peer exchanges, digital–intellectual issues in teaching research activities, leadership expectations, and evaluative orientations were investigated. Tri-dimensional data were cross-referenced through a mixed-analysis matrix, thereby enabling ecological attribution of retardant effects; subsequent results were also presented dimension by dimension along this framework.

3.2 Research Subjects and Sample

The research subjects covered teachers from primary and secondary schools, vocational colleges, and higher education institutions, as shown in Table 2. In the first phase, stratified purposive sampling was adopted, whereby 24 schools of different types across six provinces in the eastern, central, and western regions were selected. A total of 832 questionnaires were distributed, with 816 valid responses retrieved, yielding an effective response rate of 98.1%. The inclusion criteria were as follows: teaching duties were undertaken, and at least one semester of experience with intelligent education platforms had been accumulated. In the second phase, a multi-case design was employed, in which 8 teachers with information saturation and typical characteristics, along with their respective schools, were selected from the first-phase sample as units of analysis. Case selection followed the principle of maximum variation, encompassing individuals with good technological adaptation as well as those with significant retardant effects, and school ecologies with disparate resources. In the third phase, experimental and control groups were established in 2 partner schools, involving 42 teachers, and a one-semester comparative intervention of the pathway was conducted.

3.3 Data Sources and Collection Procedures

Four types of data sources were included. First, a self-developed diagnostic questionnaire on retardant effects in teachers' digital–intellectual thinking was administered, which covered dimensions such as cognitive inertia,

knowledge updating, professional motivation, and environmental support. Through pretesting and item analysis, the Cronbach's α coefficient for the overall scale was found to be 0.89, with subscale α values ranging between 0.82 and 0.91. Second, a semi-structured interview protocol was employed, which focused on teachers' interaction narratives with technologies and communities; each interview session lasted 60–90 minutes, and lesson reflection logs and teaching research records were collected simultaneously. Third, classroom observation records and back-end logs from intelligent teaching platforms were obtained, from which behavioral traces such as technology usage frequency, functional preferences, and dwell time were extracted. Fourth, school documents and policy texts were gathered, including informatization development plans and school-based teaching research institutional documents. All data were collected under informed consent; interviews were audio-recorded in full and transcribed, and platform logs were anonymized.

Table 2. Basic sample information ($n = 816$)

Characteristic Variable	Category	Proportion (%)
Gender	Male	35.7
	Female	64.3
Years of teaching	≤5 years	25.0
	6–15 years	40.0
	16–25 years	25.0
	≥26 years	10.0
	Primary	35.0
School level	Junior high	25.0
	Senior high	20.0
	Vocational/Higher education	20.0
Region	Eastern	35.0
	Central	33.0
	Western	32.0
School type	Urban high-quality	25.0
	Urban ordinary	30.0
	County-town	25.0
	Rural	20.0

3.4 Data Analysis Procedures

Quantitative data were analyzed using SPSS 27.0, through which descriptive statistics, difference tests, and hierarchical regression analyses were conducted to identify key predictive variables of retardant effects. Qualitative data were analyzed using reflexive thematic analysis, with initial coding performed back-to-back by two coders, and data management facilitated by Nvivo 14. The coding process was guided by the deductive logic of the “cognitive agent–technological environment–community of practice” framework, while remaining open to emergent themes. Constant comparison across categories was conducted until no new themes could be generated from additional cases. Reliability assurance measures included the following: an inter-coder consistency coefficient (κ) of 0.83 was achieved; preliminary interpretations were fed back to participating teachers for member checking; and thick descriptions of contextual details were provided to enhance transferability. Quantitative results and qualitative findings were cross-referenced and integrated through a mixed-analysis matrix, thereby enhancing convergent validity.

4. Research Results

Through questionnaire surveys ($n = 816$) and classroom observations, the retardant effects in the development of digital–intellectual thinking among teachers of different age groups were diagnosed. The results are presented in Table 3 and Table 4.

4.1 Cognitive Inertia Fixation

The principle of “not altering the original teaching process” was prioritized in instructional design by 72.3% of teachers when using technology. As revealed by classroom observations, the average frequency of active invocation of learning analytics dashboards by teachers was 1.2 times per class session, with 83.5% of such invocations serving attendance tracking purposes, and only 6.8% being used for cognitive diagnosis or learning analysis. Among teachers with more than 5 years of teaching experience, this fixation tendency was found to be more pronounced, with the proportion of technology use expansion being 21.4 percentage points lower than that of novice teachers.

4.2 Data–Instructional Meaning Construction Fracture

In an experimental task involving the design of remedial instruction based on error-distribution heatmap data, only 24.1% of teachers were able to propose clear instructional adjustment strategies. Of the teachers, 41.6% remained at the descriptive level of “pointing out error distributions,” and 34.3% were unable to establish any connection between data and instructional actions. As indicated by questionnaire self-reports, the proportion of teachers who endorsed the statement “I can understand the data, but I do not know what it tells me to change” reached 58.7%, which was considerably higher than the proportion who reported “inability to operate the data analysis system” (21.3%).

Table 3. Core indicators and detection rates of the four types of retardant effects

Retardant Type	Key Indicators	Data
Cognitive inertia fixation	Priority given to maintaining original teaching processes	72.3%
	Average frequency of active invocation of learning dashboard per single class session	1.2 times
	Proportion of invocations used for attendance tracking	83.5%
	Proportion of invocations used for cognitive diagnosis	6.8%
Data–meaning construction fracture	Ability to propose teaching adjustment strategies based on error data	24.1%
	Ability only to describe data distributions without linking to actions	41.6%
	Self-reported “not knowing what the data indicate should be changed”	58.7%
	Self-reported “inability to operate the data analysis system”	21.3%
Human–machine collaborative perception misalignment	Direct adoption of system scores in essay evaluation	41.8%
	Never using system suggestions in essay evaluation	37.6%
	Ability to make independent judgments in conjunction with system suggestions	20.6%
	Self-reported “not understanding” or “completely not understanding” algorithmic logic	63.2%
Professional identity narrative disruption	Mean score (5-point scale) for “Technology makes me doubt the core value of the teaching profession”	3.89
	Proportion scoring ≥ 3 on this item	68.2%
	Willingness to reduce the use of intelligent tools if conditions permit	46.5%

Table 4. Differences in retardant effects across teacher groups

Comparison Dimension	Groups	Indicator	Value	Significance
Years of teaching	>5 years vs. novice teachers	Difference in proportion of technology use expansion	21.4 percentage points lower	-
Professional type	Skills-oriented vs. theory-oriented teachers	Mean score of identity doubt	4.21 vs. 3.52	$p < 0.01$
Years of teaching	10–20 years (core teachers)	Proportion willing to reduce intelligent tool use	54.3%	-

Note: The hyphen (-) denotes not applicable.

4.3 Human–Machine Collaborative Perception Misalignment

Taking the intelligent essay scoring system as an example, teachers’ usage behaviors were found to exhibit a distinctly bipolar distribution: 41.8% of teachers “directly adopted system scores as final evaluations,” while 37.6% “never used system suggestions,” with a total of 79.4% of teachers being in a state of collaborative misalignment. The proportion of teachers able to make independent judgments in conjunction with system suggestions was only 20.6%. As revealed by in-depth interviews with misaligned teachers, 63.2% of respondents reported being “not familiar” or “completely unfamiliar” with the decision-making logic of the algorithms.

4.4 Professional Identity Narrative Disruption

On the questionnaire item “Digital–intelligent technologies make me doubt the core value of the teaching profession,” the mean score was 3.89 (on a 5-point scale), with 68.2% of respondents scoring 3 or above. This mean score was significantly higher among skills-oriented teachers than among theory-oriented teachers (4.21 vs. 3.52, $p < 0.01$). In response to the question “If conditions permit, would you be willing to reduce the use of intelligent tools?” 46.5% of teachers expressed avoidance tendencies, with this proportion being highest (54.3%) among the group with 10–20 years of teaching experience.

4.5 Key Statistical Tests

As shown in Table 5, correlation analyses revealed that all three dimensions—cognitive agent, technological environment, and community of practice—were significantly negatively correlated with retardant effects ($r = -0.52, -0.38, \text{ and } -0.41$, respectively; all $p < 0.01$). Hierarchical regression analyses indicated that, after controlling for years of teaching and school level, the three dimensions jointly explained 38% of the variance in retardant effects ($\Delta R^2 = 0.38$), with the cognitive agent demonstrating the strongest predictive power ($\beta = -0.41$), followed by community of practice ($\beta = -0.26$) and technological environment ($\beta = -0.22$); all predictors reached statistical significance ($p < 0.01$). Difference tests further confirmed that the mean score for identity narrative disruption was significantly higher among skills-oriented teachers than among theory-oriented teachers (4.21 vs. 3.52, $t = 4.86$, $p < 0.01$), and that the overall mean score of retardant effects was significantly higher among teachers with 10–20 years of teaching experience than among those with less than 5 years (3.76 vs. 3.12, $t = 5.13$, $p < 0.01$), constituting a high-risk group characterized by superimposed retardant effects.

Table 5. Main statistical test results

Test Type	Core Result	Significance
Cognitive agent–retardant correlation	$r = -0.52$	$p < 0.01$
Technological environment–retardant correlation	$r = -0.38$	$p < 0.01$
Community of practice–retardant correlation	$r = -0.41$	$p < 0.01$
Joint explanatory power of three dimensions	$\Delta R^2 = 0.38$	$p < 0.01$
Identity disruption: skills-oriented vs. theory-oriented	4.21 vs. 3.52	$t = 4.86, p < 0.01$
Retardant effects: 10–20 years vs. <5 years	3.76 vs. 3.12	$t = 5.13, p < 0.01$

4.6 Interview Corroboration of Retardant Effects

The interview findings were mutually corroborative with the quantitative results. Representative cases are presented in Table 6.

Table 6. Representative interview excerpts and quantitative correspondences for the four types of retardant effects

Retardant Type	Representative Interview Excerpt	Quantitative Correspondence
Cognitive inertia	“I already know in my mind where students will get stuck; going through the data again is just a waste of time” (18 years of teaching in the Chinese language).	The teacher invoked the dashboard only 7 times throughout the semester, all for attendance tracking; full-sample detection rate: 72.3%.
Meaning construction fracture	“The system only tells me half the class got it wrong, but it doesn’t distinguish whether it’s a carry error or a decimal-point error” (12 years of teaching in mathematics).	Belongs to the 41.6% group that “can only describe data distributions”; system operation score ranked in the top 30%.
Collaborative misalignment	“I use the system scores to satisfy inspection requirements, then manually re-grade everything and keep both sets of scores” (8 years of teaching at a vocational school).	Explains the polarization between 41.8% direct adoption and 37.6% complete neglect.
Identity disruption	“Students can generate reports with artificial intelligence more clearly than I could teach in a whole semester—what use is my experience anymore?” (22 years of teaching in accounting at a higher vocational school).	Identity doubt mean score: 4.8; mean score for skills-oriented teacher group: 4.21.

In summary, the detection rates of the four types of retardant effects, from highest to lowest, were as follows: cognitive inertia fixation (72.3%), meaning construction fracture (58.7% self-reported), identity narrative disruption (68.2% agreement), and collaborative perception misalignment (79.4% at the behavioral level). Significant differentiation was observed across different groups, with teachers having 10–20 years of teaching experience and skills-oriented teachers constituting a high-risk group characterized by superimposed retardant effects.

5. Conclusion

The retardant effects in the development of teachers’ digital–intellectual thinking are a systemic product of tri-dimensional imbalance among the cognitive agent, technological environment, and community of practice, manifesting as cognitive inertia, lagging knowledge updating, and dissipation of professional motivation. The

ecological transition pathway, guided by cognitive restructuring as a precursor, supported by environmental enabling, and safeguarded by ecological synergy, is capable of effectively breaking through the retardant cycle and promoting the adaptive development of teachers' digital–intellectual thinking. At the theoretical level, cognitive ecology theory was systematically applied to research on teachers' digital–intellectual thinking, an integrative analytical framework was constructed, and the understanding of the mechanisms underlying teachers' cognitive transformation in the digital age was deepened. At the practical level, it is recommended that teacher education institutions and schools establish an integrated support system of “diagnosis–empowerment–linkage”: regular diagnosis of digital–intellectual thinking should be conducted, cognitive scaffolds should be provided for teachers, intelligent platforms should be shifted from functional aggregation toward cognition-adaptive design, a fault-tolerant, dialogic, and shared teaching-research ecology should be fostered, and the development of digital–intellectual thinking should be incorporated into formative assessments of teacher professional growth.

In the context of the continuing acceleration of educational digital transformation, the cultivation of teachers' digital–intellectual thinking is by no means a task to be accomplished in a single day, nor can it be achieved through isolated breakthroughs. Only through holistic observation of teacher development within its situated cognitive ecology and systematic adjustment can the endogenous forces for the co-evolution of teachers and intelligent technologies be genuinely activated.

Author Contributions

Conceptualization, X.Z.Z. and R.Z.; investigation, R.Z.; writing—original draft preparation, X.Z.Z.; writing—review and editing, X.Z.Z. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- Aillerie, C., Martineaud, T., Solari Landa, M., Valle Taiman, A. C., Figueroa Iberico, A. M., & Castillo Perez, V. M. (2025). Primary school teachers' media and information literacy: A contrastive study between France and Peru. *Educ. Inf.*, 42(3), 207–222. <https://doi.org/10.1177/01678329251400648>.
- Athanassopoulos, S., Tzavara, A., Aravantinos, S., Lavidas, K., Komis, V., & Papadakis, S. (2026). Teacher education students' practices, benefits, and challenges in the use of generative AI tools in higher education. *Educ. Sci.*, 16(2), 228. <https://doi.org/10.3390/educsci16020228>.
- Çelik Çoban, A. & Büyükdereli Atadag, Y. (2026). Digital health literacy and infodemic awareness among preschool teachers: Skills, behaviors, and determinants. *Front. Public Health*, 14, 1774132. <https://doi.org/10.3389/fpubh.2026.1774132>.
- Chen, S., Salleh, S. M., & Salleh, M. A. M. (2025a). Media and information literacy among pre-service teachers: A systematic review of key trends and gaps (2013–2024). *Int. J. Inf. Manage. Data Insights*, 5(2), 100348. <https://doi.org/10.1016/j.jjime.2025.100348>.
- Chen, S., Salleh, S. M., Salleh, M. A. M., & Zhou, F. (2025b). Construction of an evaluation index system for teacher education students' information literacy: An empirical analysis of Chinese postgraduate programs. *Humanit. Soc. Sci. Commun.*, 12(1), 1185. <https://doi.org/10.1057/s41599-025-05561-w>.
- Clarke, J. H., Kretz, C., & Fena, C. (2025). “It's not scary”: A university-high school collaboration to rethink approaches to information literacy. *J. Acad. Librariansh.*, 51(6), 103146. <https://doi.org/10.1016/j.acalib.2025.103146>.
- Garrison, K. L. & Oddone, K. (2026). “I am the only one teaching it”: Australian primary teacher librarians leading information literacy. *J. Libr. Adm.*, 66(2), 138–157. <https://doi.org/10.1080/01930826.2026.2630576>.
- Hamurcu, G. C. (2026). Information and technology literacy among female and male pre-service teachers: The mediating role of media literacy. *Sage Open*, 16(1), 21582440251414733. <https://doi.org/10.1177/21582440251414733>.
- Li, J. & Xue, E. (2025). Emerging gap in teachers' information literacy and educational equality policy in China. In *Emerging Gaps in Educational Equality Policy: Insights from China* (pp. 25–45). Springer Nature Singapore. https://doi.org/10.1007/978-981-96-9525-6_2.
- Lin, K. (2026). Research on the composition and development pathways of English teachers' information literacy in the context of digital transformation. *J. Soc. Sci. Cult. Dev.*, 2(11).

- <https://doi.org/10.70767/jsscd.v2i11.897>.
- Liu, J. C., Cheung, A. C. K., & Yang, H. (2025). Do school-based ChatGPT policies bridge or widen gaps in students' computer and information literacy across gender, socioeconomic status, and immigration backgrounds. *Comput. Educ.*, 244, 105536. <https://doi.org/10.1016/j.compedu.2025.105536>.
- Omar, O. H., Warfa, A. O., & Karieny, D. (2025). Teachers digital literacy and integration of information communication technology in teaching in public primary schools in Garissa township, Kenya. *Asian J. Educ. Soc. Stud.*, 51(9), 318–327.
- Pavlounis, D., Pashby, K., & Sanchez Morales, F. (2025). Linking digital, visual, and civic literacy in an era of mis/disinformation: Canadian teachers reflect on using the Questioning Images tool. *Educ. Inq.*, 16(4), 635–652. <https://doi.org/10.1080/20004508.2023.2292828>.
- Sari, I. J. & El Islami, R. A. Z. (2025). STEM literacy-oriented bioinformatics education through the development of e-BIMO for pre-service biology teachers in the 21st century. *Discov. Educ.*, 4(1), 366. <https://doi.org/10.1007/s44217-025-00836-5>.
- Tam, A. C. F. (2025). Unpacking teacher beliefs on feedback: Shaping multifaceted teacher feedback literacy. *Assess. Eval. High. Educ.*, 50(5), 661–676. <https://doi.org/10.1080/02602938.2025.2451086>.
- Tsagari, D. & Armostis, S. (2025). Contextualizing language assessment literacy: A comparative study of teacher beliefs, practices, and training needs in Norway and Cyprus. *Educ. Sci.*, 15(7), 927. <https://doi.org/10.3390/educsci15070927>.
- Wang, Y., Xu, H., Peng, H., & Li, S. (2025). Research on the hierarchical enhancement model of information literacy for university teachers in the context of education informatization 2.0. *J. Mod. Educ. Cult.*, 2(4). <https://doi.org/10.70767/jmec.v2i4.627>.
- Xiao, W., Wang, M., & Li, X. (2026). From “Digital Novices” to “Digital Pioneers”: Profiles and influencing factors of digital literacy among pre-service special education teachers. *Teach. Teach. Educ.*, 170, 105301. <https://doi.org/10.1016/j.tate.2025.105301>.
- Xie, X., Du, C., Cheng, J., & Zhang, H. (2026). From believing to behaving: Unpacking teacher emotion as the mediator between information literacy self-efficacy and information-empowered teaching engagement. *Front. Psychol.*, 16, 1696627. <https://doi.org/10.3389/fpsyg.2025.1696627>.